direct method formula. However, if the observations are large, finding the products of the observations and their corresponding frequencies, and then adding them is not only difficult and time consuming but also has chances of errors. In such cases, we can use the Assumed Mean Method to find the arithmetic mean of grouped data.

Steps

1. Assume any value of the observations as the Mean (A). Preferably, choose the middle value.
2. Calculate the deviation $d = x - A$ for each class
3. Multiply each of the corresponding frequency '***f***' with '***d***' and obtain Σfd
4. Apply the formula $\bar{X} = A + \frac{\sum fd}{\sum f}$

Example 8.2 Find the mean for the following frequency table:

Class Interval	100-120	120-140	140-160	160-180	180-200	200-220	220-240
Frequency	10	8	4	4	3	1	2

Solution

Let Assumed mean $A = 170$

Class Interval	**Frequency** f	**Mid value** x	$d = x - A$ $d = x - 170$	fd
100-120	10	110	–60	–600
120-140	8	130	–40	–320
140-160	4	150	–20	–80
160-180	4	170	0	0
180-200	3	190	20	60
200-220	1	210	40	40
220-240	2	230	60	120
	$\sum f = 32$			$\sum fd = -780$

Mean $\bar{X} = A + \frac{\sum fd}{\sum f}$

$= 170 + \left(\frac{-780}{32}\right)$

Therefore, $\bar{X} = 170 - 24.375$

$= 145.625$

Step Deviation Method

In order to simplify the calculation, we divide the deviation by the width of class intervals (i.e. calculate $\frac{x - A}{c}$) and then multiply by c in the formula for getting the mean of the data. The formula to calculate the Arithmetic Mean is

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